

SCHOOL OF COMPUTING
SECI2143 – PSDA
SEM II – 2021/2022
Exercise (Chapter 3)

QUESTION 1

The heights, in centimetres, of the students of a Computer Science class are as follows:

150	152	154	155	156	158	158
158	161	161	162	162	163	164
164	165	166	166	166	167	168
169	170	170	171	171	172	174

- a. By grouping the data into 5 classes, build a frequency table which contains class interval (first interval is 150 – 154), class boundaries, class midpoints, frequency, relative frequency and cumulative frequency. [8 marks]

- b. Plot the data on a histogram. [2 marks]

QUESTION 2

Table 2 shows deaths caused by dengue fever in Malaysia from the year 2006 – 2016.

Table 2: Dengue fever deaths from year 2006 – 2016 in Malaysia

Year	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Deaths	92	98	112	88	134	36	35	92	215	336	237

- i) Find mean, mode, median and standard deviation (6 marks)
- ii) Number of deaths above 200 is considerably high. Determine the percentile of the deaths 215? (2 marks)

QUESTION 3

Thirty students enrolled in “Statistic Using SPSS” course. The course materials, grades and lecture notes are available to students on e-learning website where frequency of access of each student to any of the pages were recorded. Table 3 below shows frequency of access for the thirty students after one month period.

Class Interval	Frequencies
0 – 4	1
5 – 9	5
10 – 14	5
15 – 19	6
20 – 24	5
25 – 29	4
30 – 34	2
35 - 39	1
40 - 44	1

- i) Find median of the number of access. (4 marks)
- ii) Find mode of the number of access. (4 marks)

QUESTION 4

The times in seconds for 15 athletics took to run 800 meters is given Table 2.

Table 2

139	148	151	140	162
182	154	171	157	142
145	178	132	148	166

- a) Draw a stem-and-leaf diagram of the data. [3 marks]
- b) Find the median, the upper and lower quartiles and the interquartile range (IQR). [4 marks]
- c) Draw a boxplot of the data. [3 marks]

QUESTION 5

For the score of golfers in the first round of a competition given in Table 3.

Table 3

Score	70	71	72	73	74	75	76	77	78	79	80	81	82
Tally				 	 	 		 					

- a) Find the median [2 marks]
 b) Find the upper and lower quartile [3 marks]
 c) Find interquartile range [2 marks]
 d) Illustrate the data with boxplot [3 marks]

QUESTION 6

Akmal designed a game where customers pay RM2 and are given 14 balls which they have to try to get into a container which is 6 metres away. If they get more than 8 balls in the container, they win RM5. Before introducing the game, she tries it out on a sample of 50 people. The number of successes scored by each person is recorded and grouped as below.

Table 3

Score	Frequency
0 – 2	9
3 – 5	15
6 – 8	20
9 – 11	4
12 – 14	2

Calculate

- i) mode (2 marks)
 ii) median (2 marks)
 iii) mean (2 marks)